

1. Identify the objective: We want f_X . ✓
2. Realize: $f_X = F'_X$ ✓
3. Recall the definition: $F_X(x) = \mathbb{P}[X \leq x] = \mathbb{P}[g(Y_1, \dots, Y_n) \leq x]$
4. Identify the region A_x in $\mathbb{R}^n$ where

$$g(y_1, \dots, y_n) \leq x$$

for every x , i.e., express

$$A_x = \{(y_1, \dots, y_n) : g(y_1, \dots, y_n) \leq x\}$$

5. Calculate

$$F_X(x) = \int \cdots \int_{\mathbb{R}^n} \mathbf{1}_{A_x}(y_1, \dots, y_n) f_Y(y_1) \cdots f_Y(y_n) dy_1 \cdots dy_n.$$

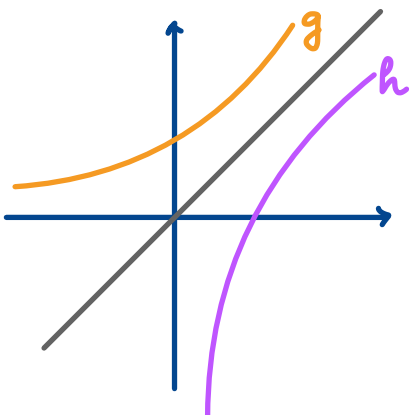
6. Differentiate: $f_X = F'_X$.
7. Pat yourself on the back!

Problem 8.3. One-to-one transformations: Step-by-step Let Y be a random variable with density f_Y . Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing differentiable function. Define $\tilde{Y} = g(Y)$. What is the density function $f_{\tilde{Y}}$ of $\tilde{Y}$ expressed in terms of f_Y and g ?

1. Identify the objective: We want $f_{\tilde{Y}}$.
2. Realize: $f_{\tilde{Y}} = F'_{\tilde{Y}}$
3. Recall the definition:

$$F_{\tilde{Y}}(x) = \mathbb{P}[\tilde{Y} \leq x] = \mathbb{P}[g(Y) \leq x]$$

4. The function g is assumed to be **strictly increasing**. In which way can you modify the inequality in the probability you obtained above to separate the random variable Y from the transformation g ?



There exists $h = g^{-1}$.
This is g 's inverse function

h is also strictly increasing

$$\begin{aligned} F_{\tilde{Y}}(x) &= \mathbb{P}[g(Y) \leq x] = \mathbb{P}[\cancel{h}(g(Y)) \leq h(x)] \\ &= \mathbb{P}[Y \leq h(x)] \end{aligned}$$

5. Express your result from above in terms of the c.d.f. F_Y of the r.v. Y .

$$F_{\tilde{Y}}(x) = \mathbb{P}[Y \leq h(x)] = F_Y(h(x))$$

6. Differentiate: $f_{\tilde{Y}} = F'_{\tilde{Y}}$.

$$\begin{aligned} f_{\tilde{Y}}(x) &= \frac{d}{dx} F_Y(h(x)) = (\text{chain rule}) \\ &= f_Y(h(x)) \cdot h'(x) \end{aligned}$$

Problem 8.4. The time T that a manufacturing distribution system is out of operation is modeled by a distribution with the following c.d.f.

$$F_T(t) = (1 - (2/t)^2) \mathbb{1}_{(2, \infty)}(t)$$

The resulting cost to the company is $Y = T^2$. Find the probability density function f_Y of the r.v. Y .

$$\rightarrow: g(t) = t^2, t > 2 \Rightarrow h(y) = \sqrt{y}, y > 4 \Rightarrow h'(y) = \frac{1}{2\sqrt{y}}, y > 4$$

$$f_T(t) = \frac{d}{dt} F_T(t) = \frac{8}{t^3} \mathbb{1}_{(2, \infty)}(t)$$

$$f_Y(y) = \frac{8}{(\sqrt{y})^3} \cdot \frac{1}{2\sqrt{y}} \mathbb{1}_{(4, \infty)}(y) = \frac{4}{y^2} \mathbb{1}_{(4, \infty)}(y) \quad \square$$

Problem 8.5. What if h is strictly decreasing? $h = g'$

$$\begin{aligned} F_{\tilde{Y}}(y) &= \mathbb{P}[\tilde{Y} \leq y] = \mathbb{P}[g(Y) \leq y] = \mathbb{P}[h(Y) \geq h(y)] \\ &= \mathbb{P}[Y \geq h(y)] = 1 - \mathbb{P}[Y \leq h(y)] = 1 - F_Y(h(y)) \end{aligned}$$

$$f_{\tilde{Y}}(y) = -f_Y(h(y)) \cdot h'(y) > 0$$

Problem 8.6. The unifying formula?

$$f_{\tilde{Y}}(y) = f_Y(h(y)) \cdot |h'(y)|$$

Do not forget: it always makes sense to simply attack a problem without giving it a "label"
Just look at the following problem:

Problem 8.7. Let T_1 and T_2 be independent shifted geometric random variables with parameters $p_1 = 1/2$ and $p_2 = 1/3$. Compute $\mathbb{E}[\min(T_1, T_2)]$.

→: $T = \min(T_1, T_2)$... counts the # of trials until the 1st success happens from either of the "coins"

⇒ $T \sim$ shifted geometric

p ... success probability for T

$p = \mathbb{P}[\text{@ least one of the "coins" is a success}]$

$p = 1 - \mathbb{P}[\text{neither "coin" is a success}]$

$$p = 1 - \underbrace{\frac{1}{2} \cdot \frac{2}{3}}_{\text{independence}} = \frac{2}{3}$$

independence

$$\mathbb{E}[T] = (?)$$

w/ $T \sim$ shifted geometric ($p = \frac{2}{3}$)

$$\mathbb{E}[\text{shifted geometric}] = 1 + \mathbb{E}[\text{geometric}]$$

$$= 1 + \frac{q}{p} = \frac{p+q}{p} = \frac{1}{p}$$

$$\mathbb{E}[T] = \frac{3}{2}$$

